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Fundamentals of Aviation and Aircraft Automation

Why start with aviation?

- Investment in human factors research for aviation began around World War I
  - No parallel effort for automotive human factors until 1970-80s*
- Aviation insights often transfer to other modes of transportation.

Curtiss JN4-D "Jenny" WWI Trainer
Curtiss Museum, Hammond, NY

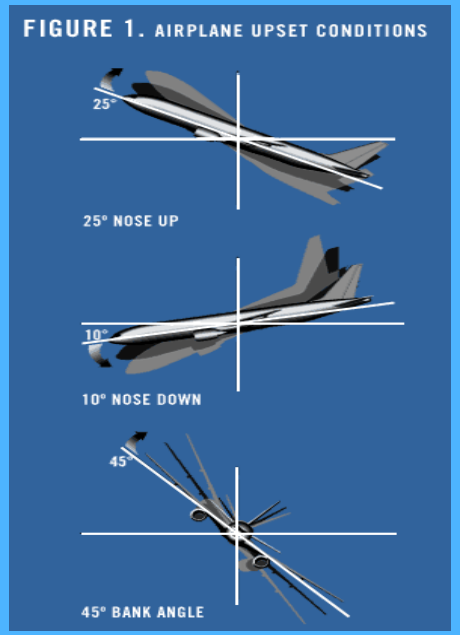
*Geels, F.W. & Penna, C.C.R (2015) Societal problems and industry reorientation: Elaborating the dialectic issue life cycle (DILC) model and a case study of car safety in the USA (1900-1995). http://dx.doi.org/10.1016/j.respol.2014.09.006

Topics

- Pilots and what they do
- Flight deck displays and controls
- Air Traffic Control
- Automated systems in aircraft
- Automation in general

What does a pilot do?

- Aviate, Navigate, Communicate
  - Maintain safe airplane configuration
  - Plan route of flight and monitor so that you get to where you're going
  - Keep Air Traffic Control and other aircraft informed



https://www.davidclark.com

Aircraft upset image from Boeing commercial aeromagazine, Aero 03; http://hibdz.skydiving.co.uk/757/767_tech/boeing/aero_03_upsets.htm

Other Pilot Tasks

- Manage flight deck systems and information, make decisions
  - Routing, speed, weather, fuel, maintenance…
- Stay alert and handle routine anomalies
  - Checklists and procedures
  - “Crew Resource Management”
- In rare cases, handle emergencies

Pilot Training and Qualifications

Student Pilot
Private Pilot
Commercial Pilot
Air Transport Pilot

Instrument Rating
Multi-engine Rating
Instructor Ratings

Seaplanes, Balloons, Gliders….

- Ground school
  - Knowledge
- In-flight and simulator training
  - Manual skills and more
- Medical evaluation

FLIGHT DECK DISPLAYS AND CONTROLS

From the 1950s to 1990s…

Round-Dial Instruments
Cessna 172

http://en.wikipedia.org/wiki/Cessna_172

By Durksteel at English Wikipedia - Own work / Originally from en.wikipedia; description page was here. Public Domain, https://commons.wikimedia.org/w/index.php?curid=2612908

The ‘90s and beyond…

Primary Flight Display

Image courtesy of Amy Pritchett

Newer Flight Decks

Cessna 172 with Garmin G1000
(https://www.garmin.com/en-US/c/aviation/flight-decks/)

"Information Automation"

http://www.sunstateaviation.com/accelppl172.html

Boeing 787

widebodyaircraft.nl

Fundamentals: How to Control an Airplane

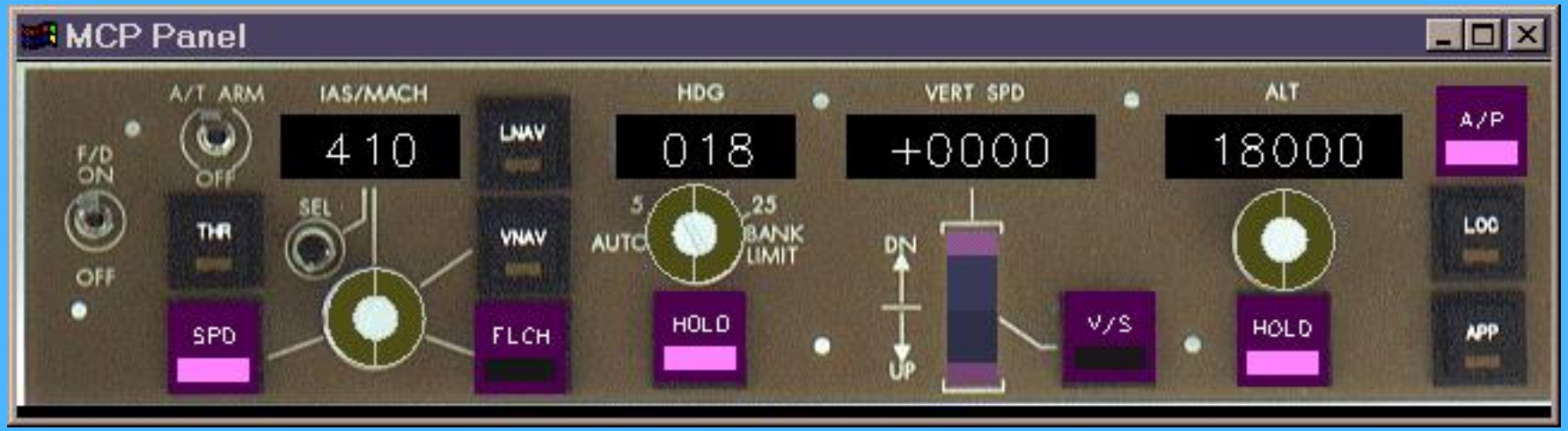
- Manually
- By setting parameter(s)
  - Vertical speed, altitude, heading…
- Flight Management Computer
  - Pre-programmed routes

How to “Aviate”: The Basics

- https://youtu.be/WhQ8Ai4fa_Q

- Flight control surfaces video by Embry Riddle Aeronautical University (6.5 min)

Flight Management
Sample Mode Control Panel



Courtesy of Amy Pritchett, Penn State University

Autoflight Systems – Control Display Unit

- Used to program the Flight Management Computer

Courtesy of Amy Pritchett, Penn State University

Volpe’s B-737 Research Simulator

AIR TRAFFIC CONTROL (ATC)

ATC Responsibilities

Joint Order (JO) 7110.65 (on FAA website)

2-1-1. ATC SERVICE

a. The primary purpose of the ATC system is to prevent a collision involving aircraft operating in the system.

b. In addition to its primary purpose, the ATC system also:

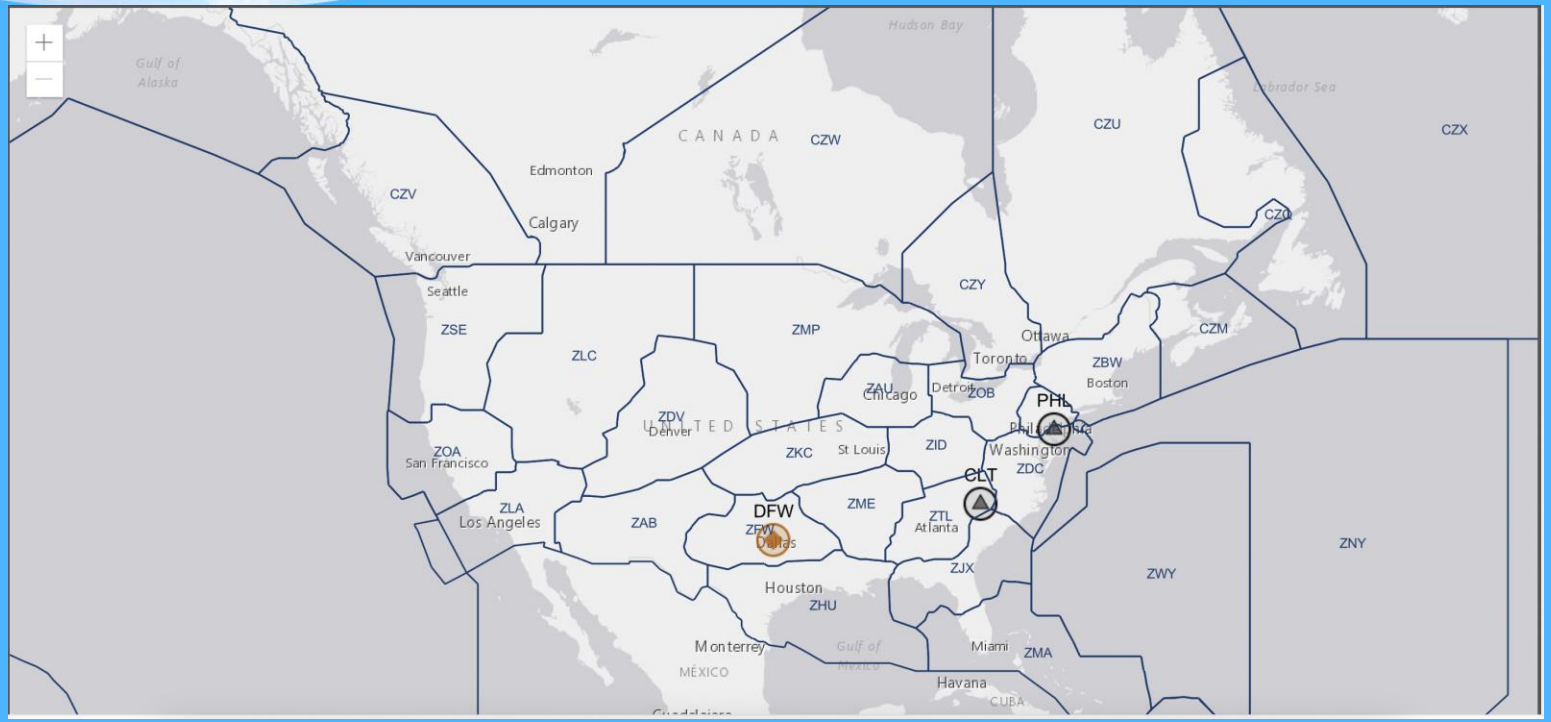
1. Provides a safe, orderly, and expeditious flow of air traffic.
2. Supports National Security and Homeland Defense missions.

c. The ATC system must provide certain additional services to the extent permitted. The provision of additional services is not optional on the part of the controller, but rather required when the work situation permits….

Air Traffic Control Facilities

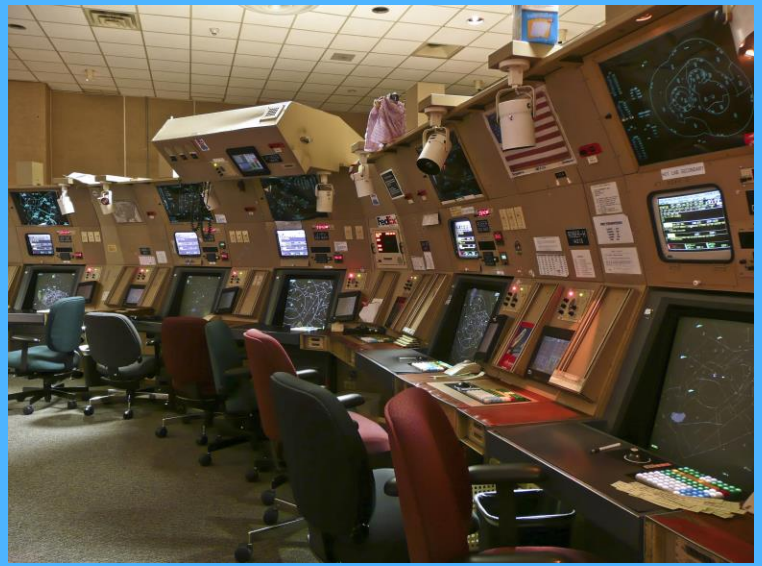
- Ground
  - Taxiing, movement on the airport surface
- Tower
  - Takeoffs and landings
- Terminal Radar Control (TRACON)
  - Within approximately 30 NM of airport
- Air Route Traffic Control Center (ARCTCC, aka Center)
- Flow Control
  - https://nasstatus.faa.gov

ARTCC (Center) Map from NAS Status Website



Air Traffic Control

New York TRACON



Newark Tower

Courtesy of Jim Kuchar, MIT Lincoln Laboratory

AUTOMATED SYSTEMS IN AIRCRAFT

Some Flight Deck Automated Systems

- Simple early system, wing-leveler (holds heading)
- 1980s and beyond: Flight path management
  - Lateral Navigation, Vertical Navigation, and more
- Power management (auto-throttle)
- Fly-by-wire (aircraft configuration)
- Auto-land
- Auto-brakes
- TCAS-Traffic Alert and Collision Avoidance System
- And more…especially in terms of information automation

Known Consequence of Flight Deck Automation

- Automation *changes and shifts* workload, it does not necessarily *reduce* workload
- For flight path management, for example
  - More preparation pre-flight
  - More workload and head-down time to enter routing changes; especially high workload under time pressure

Example: Flying in Controlled Airspace

- Pilots are used to a fairly complex operating environment in controlled airspace
  - Many commands may be given
  - Many constraints and subtleties
- However, pilots often then need to translate these into a machine that represents things very differently!
  - For example, ‘hold left at MACEY on the 045 radial, once at MACEY descend to 15,000’ and slow to 250 knots’ --, this could take 40-50 key presses on 6 different ‘pages’ of the flight deck computer interface!

Some Potential Consequences of Flight Deck Automation

- Boredom, complacency, lack of appropriate response in an emergency
- Misunderstanding or lack of knowledge of what automation is doing/considering
- Error in data entry/interaction.
- Overconfidence in automation
- False alarms
- Mode confusion
  - e.g., managing altitude by setting a descent rate in flight path angle (3.3) vs. vertical speed in hundreds of feet per minute (33) [Air Inter 148, 1992, accident]

Mitigating Risk: Evaluations Prior to Field Use

- System “certification”
  - Does “the box” work as intended?
  - Does it cause problems for other boxes?
- Pilot “approval”
  - Training and qualifications
  - Standard operating procedures
    - Normal, Abnormal, Emergency situations
  - Authorization
    - Can the crew turn on the box and use the functions?
    - What are they allowed to do if there is a failure? Partial failure?

AUTOMATION IN GENERAL

What is “automation”? Intuitive Observations

- Technology performing functions otherwise performed by people
  - Automation often is associated with fancy displays and complex input
    - Displays to indicate output, keyboards or control panels for complex interactions
  - “Glass displays” are often at the front end, but there can be automation underneath, regardless of display type
- People manage automation
  - Normal, abnormal, emergency situations
  - Partial and full failures

More Specific Definition of “Automation”

- “Automation is the automatically controlled operation of an apparatus, a process, or a system by mechanical or electronic devices that take the place of human organs of observation, decision, and effort.” (Sheridan, 1992)
- Automation may take over all or any of the evaluation, decision making and execution
  - If the automation takes over part of the task, we may call it a decision aid, advanced display, etc.
  - If the automation takes over all of the task, the operator becomes a supervisory controller, who monitors the system and changes the automation parameters.

Why automate?

- Reduce workload
  - Car cruise control, wing-leveler, heading hold
  - Offload “boring” tasks
- Improve safety
  - Perform tasks that are more ‘difficult’ or dangerous for humans
- Extend human capabilities
  - More efficiency, precision, flexibility
- Reduce costs
  - Reduce staff…e.g., flight engineer
- Because we can…?

Evolving Automation Philosophies

- The automated system has final authority.
  - Keep the pilot from doing something foolish.
- The pilot has final authority.
  - Keep the automation from doing something foolish.
- Something in between?
  - Adaptive or adaptable automation?
  - Sharing authority?
  - Team work?

Automated Systems

From Kathy Abbott, FAA

Old View

- Automation
- Give the human operator what s/he does best, give the automation what it does best
- Automation causes degradation of basic skills
- Automation should be another “crewmember”
- Automation policy
- More automation reduces risk

New View

- Automated systems
- Human-system integration to enable the human operator
- Lack of practice causes degradation of basic skills
- Automated systems are tools to help the responsible human
- Flight path management policy
- More automation introduces different risks

Managing Automated Systems Example Topics

- When/how to turn on or off
- Mode awareness
- Dealing with anomalies or confusion
- Guidance versus control

Vertical Navigation (VNAV) Example

- Pilots make choices about how and when to use Vertical Navigation (VNAV)
  - Some choose not to couple the autopilot.
    - The pilot might be better at flying the vertical path if the autopilot has a lag. (Challenger)
  - Operating VNAV creates setup and monitoring work. It does not fully relieve the pilot.

"Automated flight is great, but it's like trying to use cruise control in traffic. At some point it's better just to fly the plane." (757/67)

"Not worth setting up VNAV for Boston (BLZZR)." (G200 and E-190)

Recent Research

- Artificial Intelligence (AI)
  - Advances in sensor technology, computing power, and training databases
  - Still working on “understanding” (reasoning about representations of the world) and speed/power requirements.
  - GenAI – enhanced automation?
- Autonomy
  - e.g., a system that uses AI to make decisions where humans generally cannot intervene
- Automation research “hot topics”
  - Brittleness and Resilience
  - Monitoring and Trust
  - Human-Automation “Teaming”

From 2011…

Advantages & Disadvantages

16.400/453

Advantages	Disadvantages
• Increased capacity & productivity • Reduction of manual workload • Reduction of fatigue • Error prevention • Precision • Economy of design space	• Dehumanizing • Complacency • False alarms • Increase in mental workload • Automation-induced failures
Questionable	**Unknown**
• Overall workload reduction? • Operational cost reduced? • Training requirements reduced? • Reduction in crew size?	• Maintenance • Hardware & software costs • Redundancy • Liability issues • Personnel issues

How do these look today?

There are still legal and other unknowns.

What else?

Thoughts About Automation

- Capabilities will continue to improve
- Use of automation/autonomy is intended to reduce cost and staffing and improve safety, but the opposite can occur
- Moving from demonstration to real world is challenging
- Creeping complexity
- Exhaustive testing of complex systems is not possible

EXTRAS

Some Online Resources to Learn More about Aviation and Human Factors

- https://skybrary.aero
- https://www.faa.gov
- Dynamic Regulatory System
  - https://www.faa.gov/about/office_org/headquarters_offices/avs/programs/drs
  - https://drs.faa.gov/browse
- NASA Aviation Safety Reporting System
  - https://asrs.arc.nasa.gov
- Lessons learned from civil aviation accidents
  - https://lessonslearned.faa.gov
- https://www.airnav.com or https://skyvector.com for airport information and charts
- Real-time flow of air traffic
  - https://nasstatus.faa.gov

Deeper Dives into Aviation Automation

- *Automation Airmanship*, by Lutat & Swah
  - A book of techniques for pilots to learn, use, and manage flight deck systems
- *Human-Automation Teaming: Unintended Impacts and Mitigations for Degraded NextGen Operations,* by Smith & Baumann, 2019 http://rosap.ntl.bts.gov/view/dot/43777
- *Final report of the Performance-based operations Aviation Rulemaking Committee/Commercial Aviation Safety Team Flight Deck Automation Working Group.* (2013), available at www.faa.gov

Mitigating Risk: Operational Complexity in Flight Operations

Operational Complexity
- ATC interventions
- Aircraft equipment or performance
- Crew factors
- Operator factors
- Environment factors

ATC Intervention (such as)
- (Late) route amendments
- Unpublished restrictions
- Vectors

Operator Factors
- Independence vs. dependence on Dispatch
- Clarity and consistency of pilot roles in reviewing route

Aircraft Factors
- Lack or unreliability of automated systems
- Performance characteristics

Environment Factors
- Terrain
- Traffic
- Weather
- Prohibited airspace

Crew Factors
- Expectations
- Fatigue
- Communication style
- Distractions
- Local area familiarity
- Familiarity with different types of routes

“Mitigations”
How to address human factors issues

Individual

- Selection and training (Knowledge, Skills, and Abilities)
- Motivation, arousal (optimize attention)

Organization

- Policies and procedures (e.g., checklists)
- Safety culture

System Design

- Controls (soft or hard), “Usability” (e.g., visibility of system state, interaction with data)
- Automated systems? (take the operator out of the system)

Operational Risk

- Variation across each operation (e.g., across flights)
- Work as imagined (theory) vs. work as done (reality)

Driving vs. Flying

Driving

- Relatively little training or currency checks
- Wider population of operators
- Could stop and reset/review
- “Operational design domain” (ODD)
- Safety? (Private vs. Public risk)

And what else?

Flying

- Intensive training and currency reviews
- Relatively limited set of operators
- Certifications and approval of both pilots and aircraft
- ATC monitors flights and can aid in emergencies
- Always in motion
- Public expects high safety levels for commercial operations